FOUNDATIONS

Architecture

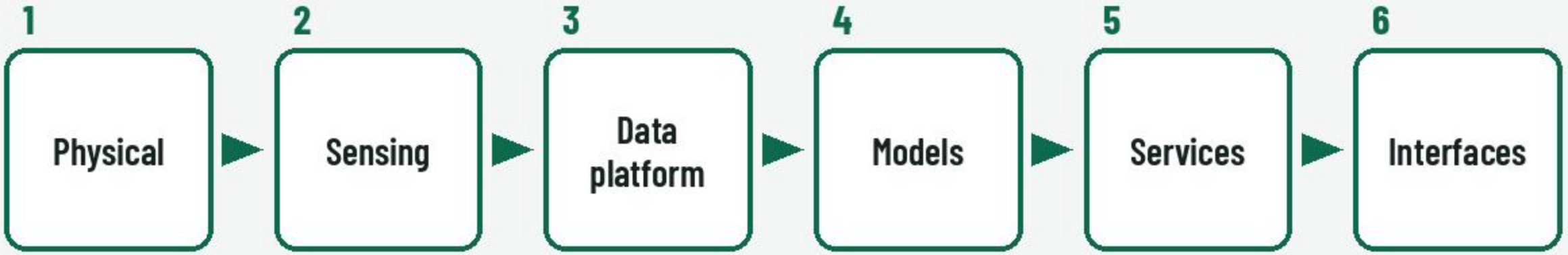
Module 2 · How the parts of a twin fit together

Transport Twin Study Path

A cyber-physical system

- Computation tightly coupled to a physical process
- Sensing, communication, control in one loop
- Timing and safety matter as much as accuracy

Layered architecture



Edge or cloud?

At the edge

Video analytics

Millisecond safety logic

Works if the network fails

In the cloud

Long history

Heavy simulation

City-wide analytics

Co-simulation and the digital thread

- Couple specialised simulators: traffic, vehicles, networks
- Standards such as FMI exchange models
- The digital thread links design, build and operation data

Five design questions

Purpose · fidelity · rate · latency · owner

Answer them before choosing any product